

INVITED REVIEW PAPER

A REVIEW OF BOOTSTRAP METHODS

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Summary: A survey of some developments in the nonparametric bootstrap methodology is given, concentrating on basic ideas and applications rather than theoretical considerations. Topics include statistical error, confidence intervals, double bootstrapping, smoothed and modified bootstrap, and bootstrap adaptive statistical procedures. A new point estimation technique which is based on approximating functionals and on the bootstrap, is proposed and discussed.

1. INTRODUCTION

Efron (1979) introduced a very general resampling procedure for estimating the distributions of statistics based on independent observations. This resampling procedure he called the bootstrap. The procedure is more widely applicable and perhaps has a sounder theoretical basis than the popular Quenouille-Tukey jackknife. Efron (1979, 1982) considered a number of statistical problems and demonstrated the feasibility of the bootstrap method. The bootstrap has some attractive properties for the statistical practitioner: it requires few assumptions, little modelling or analysis, and can be applied in an automatic way in a wide variety of situations, regardless of their theoretical complexity. The bootstrap can routinely answer questions which are too complicated for traditional statistical analysis.

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The aim of the present paper is to review and illustrate many of the developments in bootstrap methodology, and to highlight key ideas and the scope of the technique. The material in this paper is discussed in the context of the nonparametric bootstrap, since the bootstrap is used most often in that circumstance. The choice of material inevitably reflects personal interests, however, so that the paper is in no way a comprehensive review. The presentation is at a relaxed mathematical level, omitting most proofs, regularity conditions, and technical details.

2. THE BOOTSTRAP METHOD

A formal description of the bootstrap is the following: Let $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ be a sample of independent, identically distributed (i.i.d.) random variables with common distribution function F and suppose $T_n(\mathbf{X}_n; F)$ is some specified random variable of interest, possibly depending upon the unknown distribution F . Let F_n denote the empirical distribution function (e.d.f.) of $\mathbf{X}_n$, that is, the distribution with mass n^{-1} at each of the points $X_1, X_2, \dots, X_n$. The bootstrap method consists of approximating the sampling distribution of $T_n(\mathbf{X}_n; F)$ under F by the bootstrap distribution of $T_n(\mathbf{X}_n^*; F_n)$ under F_n ; where $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$ denotes a random sample of size n from F_n . The difficult part of the bootstrap procedure is the actual calculation of the bootstrap distribution. Efron (1979) suggests a Monte Carlo approximation: repeated realizations of $\mathbf{X}_n^*$ are generated by taking random samples of size n from F_n , say $\mathbf{X}_n^*(1), \dots, \mathbf{X}_n^*(B)$, and the sampling distribution of the corresponding values $T_n(\mathbf{X}_n^*(1); F_n), \dots, T_n(\mathbf{X}_n^*(B); F_n)$ is taken as an approximation to the actual bootstrap distribution of $T_n(\mathbf{X}_n^*; F_n)$. This approximation can be made arbitrarily accurate by taking B sufficiently large. In some cases it is possible to give an explicit expression for the bootstrap estimator and consequently Monte Carlo approximation is not always required. A trivial example is given in Section 3.2, and a nontrivial example is the bootstrap estimate of standard error of the median discussed by Efron (1979).

The bootstrap is therefore a computer-based method, which substitutes considerable amounts of computation for theoretical analysis. Even for relatively simple problems, computer-intensive methods like the bootstrap

become increasingly feasible and popular in an era of fast computers and low computational costs. "This is not an argument against theory, of course, only against unnecessary theory. Most common statistical methods were developed in the 1920s and 1930s, when computation was slow and expensive. Now that computation is fast and cheap we can hope for and expect changes in statistical methodology", (Efron and Tibshirani, 1986).

3. ESTIMATION OF STATISTICAL ERROR

To get a better understanding of the bootstrap we first consider the following example:

3.1 EXAMPLE: ESTIMATION OF PROBABILITIES

Suppose we wish to estimate

$$p_1 = P(\sqrt{n}(\bar{X}_n - \mu) \leq 0, 5)$$

where $\mu = E(X_i)$ and $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

The bootstrap can be used to solve this problem as follows:

1. Having observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$, construct the empirical distribution function F_n ,

$$F_n : \text{mass } n^{-1} \text{ on } x_i, i = 1, 2, \dots, n$$

2. Draw a "bootstrap random sample" $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$ from F_n . In other words, each X_i^* is drawn independently with replacement and with equal probability n^{-1} from the set $\{x_1, x_2, \dots, x_n\}$. Calculate

$$T_n(\mathbf{X}_n^*; F_n) = \sqrt{n}(\bar{X}_n^* - \bar{X}_n)$$

where $\bar{X}_n^* = n^{-1} \sum_{i=1}^n X_i^*$.

3. Independently repeat step 2 some number "B" times, obtaining bootstrap replications $T_n(\mathbf{X}_n^*(1); F_n), \dots, T_n(\mathbf{X}_n^*(B); F_n)$, and calculate

$$\hat{p}_1 = B^{-1} \sum_{b=1}^B I(T_n(\mathbf{X}_n^*(b); F_n) \leq 0, 5)$$

where $I(A)$ is the indicator function of the event A . As $B \rightarrow \infty$, $\hat{p}_1$ converges to the bootstrap estimator of p_1 . Later we will discuss how large B needs to be taken.

If we wish to calculate estimators $\hat{p}_2$ and $\hat{p}_3$ for

$$p_2 = P(\sqrt{n}(\bar{X}_n - \mu)/\sigma \leq 0,5)$$

$$p_3 = P(\sqrt{n}(\bar{X}_n - \mu)/s_n \leq 0,5)$$

where $\sigma^2 = \text{Var}(X_i)$ and $s_n^2 = n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, we follow steps 1-3 exactly as above except now using

$$T_n(\mathbf{X}_n^*; F_n) = \sqrt{n}(\bar{X}_n^* - \bar{X}_n)/s_n$$

and

$$T_n(\mathbf{X}_n^*; F_n) = \sqrt{n}(\bar{X}_n^* - \bar{X}_n)/s_n^*$$

respectively, where s_n^* is the bootstrap sample standard deviation.

How accurate are these bootstrap estimators? The results of a Monte Carlo study based on $B = 1000$ bootstrap replications in each of 1000 Monte Carlo trials are shown in Tables 1 and 2. Here "trial" refers to a new choice of the data $X_1, X_2, \dots, X_n$ from F . Two sample sizes, $n = 10, 20$, were taken and the distributions considered were the standard normal, $F \sim N(0, 1)$, and standard exponential with parameter 1, $F \sim \text{Exp}(1)$. The tables display Monte Carlo estimates of $E(\hat{p}_i)$, $i = 1, 2$, and root mean squared errors, denoted by "Average" and $\sqrt{MSE}$ respectively. Results for $\hat{p}_3$ are omitted from the tables, since they are similar to those of $\hat{p}_2$.

The results in Tables 1 and 2 show that $\hat{p}_2$ is nearly unbiased, especially in the normal case. Also, for each fixed n , $\hat{p}_2$ performs better than $\hat{p}_1$ for estimating $p_1 = p_2$ (since $\sigma = 1$), in that its root mean squared error is always much lower. This can be ascribed to the fact that $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ is an asymptotically pivotal statistic, that is, its limit distribution (as $n \rightarrow \infty$) does not depend on unknown quantities, whereas $\sqrt{n}(\bar{X}_n - \mu)$ is non-pivotal. As a matter of fact, Singh (1981) proved that under certain conditions the following results hold with probability one as $n \rightarrow \infty$

$$(1) \hat{p}_1 = p_1 + O(n^{-\frac{1}{2}})$$

$$(2) \hat{p}_2 = p_2 + O(n^{-1}).$$

Table 1
Estimates of probabilities: $F \sim N(0, 1)$

	p_1	Average	$\sqrt{MSE}$
n=10	0,691	0,713	0,054
n=20	0,691	0,703	0,037

	p_2	Average	$\sqrt{MSE}$
n=10	0,691	0,690	0,017
n=20	0,691	0,691	0,015

Table 2
Estimates of probabilities: $F \sim Exp(1)$

	p_1	Average	$\sqrt{MSE}$
n=10	0,719	0,749	0,080
n=20	0,712	0,729	0,055

	p_2	Average	$\sqrt{MSE}$
n=10	0,719	0,701	0,025
n=20	0,712	0,703	0,018

The advantages of pivoting when using the bootstrap for Euclidean-data problems are becoming widely known; see Hinkley and Wei (1984), Hartigan (1986), Beran (1987), and Hall (1987, 1988a,b).

3.2 THE BOOTSTRAP ESTIMATE OF STANDARD ERROR

We now present an exposition of the bootstrap estimate of standard error for one-sample situations. Suppose $\hat{\theta} = \hat{\theta}(X_1, X_2, \dots, X_n)$ is a real-valued statistic and let $\sigma(F)$ indicate the standard error of $\hat{\theta}$,

$$\sigma(F) = [Var_F \hat{\theta}]^{\frac{1}{2}}$$

Of course $\sigma(F)$ is also a function of the sample size n and the form of the statistic $\hat{\theta}$, but since both of these are known they need not be indicated in the notation. The bootstrap estimate of standard error is

$$(3.1) \quad \hat{\sigma} = \sigma(F_n) = [Var_{\star} \hat{\theta}^{\star}]^{\frac{1}{2}}$$

where F_n is the empirical distribution, and $\hat{\theta}^* = \hat{\theta}(X_1^*, X_2^*, \dots, X_n^*)$ with $X_1^*, X_2^*, \dots, X_n^*$ i.i.d. random variables (rv's) from F_n . Var_* denotes variance over the conditional law of X_n^* given X_n .

As a simple example, suppose $\hat{\theta} = \bar{X}_n$, in which case $\sigma(F) = [\mu_2(F)/n]^{\frac{1}{2}}$, where

$$\mu_2(F) = \int_{-\infty}^{+\infty} (x - E_F(X))^2 dF(x)$$

Note that $\hat{\sigma} = [s_n^2/n]^{\frac{1}{2}}$, so that expression (3.1) can be written in closed form, in that case.

In most cases there is no simple expression for the function $\sigma(F)$. Nevertheless, it is easy to evaluate $\hat{\sigma} = [Var_* \hat{\theta}^*]^{\frac{1}{2}}$ numerically by means of the following Monte Carlo algorithm:

1. Construct F_n .
2. Draw a bootstrap random sample $X_1^*, X_2^*, \dots, X_n^*$ from F_n and calculate $\hat{\theta}^* = \hat{\theta}(X_1^*, X_2^*, \dots, X_n^*)$.
3. Independently repeat step 2 some number "B" times, obtaining bootstrap replications $\hat{\theta}^*(1), \hat{\theta}^*(2), \dots, \hat{\theta}^*(B)$, and calculate

$$\hat{\theta}^*(\cdot) = B^{-1} \sum_{b=1}^B \hat{\theta}^*(b)$$

$$\hat{\sigma}_B = [(B-1)^{-1} \sum_{b=1}^B \{\hat{\theta}^*(b) - \hat{\theta}^*(\cdot)\}^2]^{\frac{1}{2}}$$

It is easy to see that as $B \rightarrow \infty$, $\hat{\sigma}_B \rightarrow \hat{\sigma} = \sigma(F_n)$, the bootstrap estimate of standard error. For most situations B in the range 50 to 200 is quite adequate for estimating standard errors. However, for other bootstrap calculations, e.g. for setting bootstrap confidence intervals, larger values of B are required.

How well does the bootstrap work? Table 3 gives the answer in one situation. Here the statistic $\hat{\theta}$ of interest was the 25 % trimmed mean, i.e. the average of the central 50 % of the sample. Two distributions F were investigated, a standard normal, $F \sim N(0, 1)$, and a standard negative exponential, $F \sim Exp(1)$. The sample size was $n = 15$, and $\hat{\sigma}_B$

was based on $B = 500$ bootstrap replications in each of 1000 Monte Carlo trials. Summary statistics are given in Table 3, both for $\hat{\sigma}_B$ and Tukey's jackknife estimate of standard error

$$\hat{\sigma}_J = [((n-1)/n) \sum_{i=1}^n \{\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)}\}^2]^{\frac{1}{2}}$$

where

$$\hat{\theta}_{(i)} = \hat{\theta}(X_1, X_2, \dots, X_{i-1}, X_{i+1}, \dots, X_n), \quad \hat{\theta}_{(\cdot)} = n^{-1} \sum_{i=1}^n \hat{\theta}_{(i)}.$$

Both the bootstrap and the jackknife were nearly unbiased. The bootstrap performed better in the sense that its root mean squared error was lower. Efron and Tibshirani (1986) presented similar results in their Table 1. Our results differ slightly from theirs since they based their experiments on $B = 200$ bootstrap replications in each of 200 Monte Carlo trials.

Table 3

A sampling experiment comparing the bootstrap and jackknife estimates of standard error for the 25 % trimmed mean; sample size $n=15$

	$F \sim N(0,1)$		$F \sim Exp(1)$	
	Ave	$\sqrt{MSE}$	Ave	$\sqrt{MSE}$
Bootstrap $\hat{\sigma}_B$:	0,285	0,071	0,242	0,079
Jackknife $\hat{\sigma}_J$:	0,282	0,094	0,230	0,091
True $\sigma(F)$:	0,283		0,237	

4. BOOTSTRAPPING COMPLICATED DATA SETS

The bootstrap can be applied to much more complicated situations. A regression model is a familiar example of a complicated data structure. We observe $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$, where

$$X_i = g(\beta; \mathbf{t}_i) + \epsilon_i, \quad i = 1, 2, \dots, n.$$

Here β is a vector of unknown parameters that we wish to estimate; for each i , $\mathbf{t}_i$ is an observed vector of covariates and g is a known function. The ϵ_i 's are an i.i.d. sample from some unknown distribution F with

$E(\epsilon_i) = 0$. Having observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$, we use some fitting technique to estimate β , perhaps least squares,

$$\hat{\beta} : \min_{\beta} \sum_{i=1}^n [x_i - g(\beta; t_i)]^2$$

and wish to say something about the sampling distribution of $\hat{\beta}$. The bootstrap can be applied as follows:

1. Construct the empirical distribution $\hat{F}_n$ of the residuals

$$\hat{\epsilon}_i = x_i - g(\hat{\beta}; t_i), \quad i = 1, 2, \dots, n.$$

2. The bootstrap sample is

$$X_i^* = g(\hat{\beta}; t_i) + \epsilon_i^*$$

where $\epsilon_1^*, \epsilon_2^*, \dots, \epsilon_n^*$ are i.i.d. rv's taken from $\hat{F}_n$. Notice that the ϵ_i^* are independent bootstrap rv's, even though the $\hat{\epsilon}_i$ are not independent rv's in the usual sense.

3. Calculate

$$\hat{\beta}^* : \min_{\beta} \sum_{i=1}^n [x_i^* - g(\beta; t_i)]^2$$

where $x_1^*, x_2^*, \dots, x_n^*$ are the realization of $X_1^*, X_2^*, \dots, X_n^*$ respectively, and approximate the sampling distribution of $\hat{\beta}$ by the bootstrap distribution of $\hat{\beta}^*$.

For example, consider the estimation of $E(\hat{\beta}) - \beta$, the bias of $\hat{\beta}$. The bootstrap estimate is simply

$$E_*(\hat{\beta}^*) - \hat{\beta},$$

which can be approximated by Monte Carlo methods: repeat steps 2 and 3 above "B" times independently, obtaining bootstrap replications $\hat{\beta}^*(1), \hat{\beta}^*(2), \dots, \hat{\beta}^*(B)$, and calculate

$$B^{-1} \sum_{b=1}^B \hat{\beta}^*(b) - \hat{\beta}.$$

In many applications the form of the function g is not known and also needs to be determined using the data. For example, in polynomial regression the degree of the polynomial to be used needs to be selected. The bootstrap methodology can also be used to solve this (and other) model selection problems – see, e.g., Linhart and Zucchini (1986).

Freedman (1981) and Wu (1986) provide a detailed discussion of bootstrap applications to regression models. Navidi (1989) uses Edgeworth expansions to examine the accuracy of bootstrap estimates of the distributions of linear combinations of regression parameter estimates. It turns out that asymptotically, the bootstrap is better than the normal approximation when the elements of the design matrix increase without bound. In other situations, the bootstrap is always at least as good as the normal approximation. Efron and Tibshirani (1986) discuss several interesting examples of the application of the bootstrap to some quite complicated statistical procedures and data structures such as time series and censored data. Also, De Beer and Swanepoel (1989) show how the bootstrap can be applied to obtain a modified Durbin-Watson test for serial correlation in a general regression model under departures from the assumption of normality.

5. THE SMOOTHED BOOTSTRAP

So far, the empirical distribution function F_n was used in the bootstrap calculations. This is certainly not a bad choice. It is *asymptotically minimax* and is the *nonparametric maximum likelihood estimator* of F (Beran, 1984). However, since F_n is a discrete distribution, samples constructed from F_n in the bootstrap simulation will have some rather peculiar properties. All the values taken by the members of the bootstrap samples will be drawn from the original sample values, and nearly every sample will contain repeated values. This seems unacceptable when dealing with continuous variables. The smoothed bootstrap is a modification to the bootstrap procedure to avoid bootstrap samples with these properties. The essential idea of the smoothed bootstrap is to perform the repeated sampling not from F_n itself, but from a smoothed version of F_n such as a

kernel estimator of F which is defined by

$$F_{n,c}(x) = n^{-1} \sum_{i=1}^n K((x - X_i)/c)$$

where $c = c_n$ is a sequence of smoothing parameters such that $c_n \rightarrow 0$ as $n \rightarrow \infty$, and K is a known continuous cumulative distribution function (c.d.f.) symmetric around zero. Azzalini (1981) obtained second order results that show an asymptotic (as $n \rightarrow \infty$) improvement in the estimation of F by using $F_{n,c}$ instead of the e.d.f. F_n provided certain regularity conditions on F are met and that the sequence c_n converges to 0 at a certain rate.

From a computational point of view, it is easy to see that a bootstrap sample $X_1^*, X_2^*, \dots, X_n^*$ from $F_{n,c}$ can be obtained in a simple way: if $Y_1^*, Y_2^*, \dots, Y_n^*$ are independent rv's with common c.d.f. F_n and if $V_1, V_2, \dots, V_n$ are independent rv's with common c.d.f. K and if these two samples are independent, then we may take

$$X_i^* = Y_i^* + cV_i, \quad i = 1, 2, \dots, n.$$

The use of an estimator like $F_{n,c}$ in the bootstrap, or a variation thereof, was first suggested in the seminal paper of Efron (1979). He recognized that for certain statistics, the discreteness of the bootstrap distribution based on the empirical distribution function might lead to problems. The question of whether the smoothed bootstrap is superior to the usual bootstrap, and for which smoothing parameter c , has been a research topic. However, in most cases, no definitive answers exist, especially in small samples. Silverman and Young (1987) showed that in estimating a functional $T(F)$, there exists an unknown smoothing parameter c for which the (finite sample) mean squared error of the smoothed bootstrap $T(F_{n,c})$ is smaller than that of the usual bootstrap estimate $T(F_n)$, provided that certain conditions on both T and the unknown distribution function F are satisfied. Note that this result does not specify the value of the smoothing parameter c for which the smoothed bootstrap is superior, so that it is not very useful in practice. Hall, DiCiccio, and Romano (1989) showed that in estimating the variance of a sample quantile, the rate of convergence of

the relative error can be improved by using a smoothed bootstrap instead of the usual bootstrap. Finally, Romano (1988a) showed that if the functional of interest can be viewed as a functional of the density, bootstrap methods can even be inconsistent unless the resampling is done from a sufficiently smooth estimate of the distribution function. There is still a lot of work to be done in this area, especially for small samples. A result in the spirit of the work of Silverman and Young (1987) which gives an explicit value of the smoothing parameter c would certainly be welcomed.

6. CONFIDENCE INTERVALS AND BANDS

6.1 CONFIDENCE INTERVALS

The purpose of obtaining $\hat{\sigma}$, the estimated standard error of an estimator $\hat{\theta}$ of an unknown parameter θ , is often to set approximate (when F is unknown) $(1 - \alpha)$ -confidence intervals for θ . These standard intervals are typically of the crude form

$$\hat{\theta} \pm z(\alpha/2)\hat{\sigma}$$

where $z(\alpha/2)$ is the $100(1 - \alpha/2)$ percentile point of a standard normal distribution, and $\hat{\sigma}$ is either the bootstrap estimate $\hat{\sigma}_B$ or the jackknife estimate $\hat{\sigma}_J$ as defined above.

However, these standard intervals for θ can often be quite inaccurate, in the sense that the nominal coverage probability is not attained. The bootstrap can be used to produce nonparametric confidence intervals which perform better. An almost bewildering array of bootstrap methods for constructing confidence intervals exists in the literature. DiCiccio and Romano (1988) and Hall (1988a) give exhaustive surveys. A short discussion of the following nonparametric procedures will now be given:

1. bootstrap- t
2. pivotal iteration
3. percentile, bias-corrected percentile, accelerated bias-corrected percentile.

1. The bootstrap- t

Suppose $a = a(F)$ is a constant satisfying

$$P(|\hat{\theta} - \theta|/\hat{\sigma} \leq a) \doteq 1 - \alpha$$

from which it is evident that if $a(F)$ were known a $(1 - \alpha)$ -confidence interval for θ would be

$$\hat{\theta} \pm a\hat{\sigma}.$$

The obvious suggestion now is to replace $a(F)$ by the bootstrap estimate $a(F_n)$; thus, an approximate $(1 - \alpha)$ symmetric confidence interval for θ is

$$\hat{\theta} \pm a(F_n)\hat{\sigma}$$

where $a(F_n)$ is defined by

$$P^*(|\hat{\theta}^* - \hat{\theta}|/\hat{\sigma}^* \leq a(F_n)) = 1 - \alpha$$

and $\hat{\theta}^*, \hat{\sigma}^*$ are the estimates $\hat{\theta}, \hat{\sigma}$ based on a bootstrap random sample $X_1^*, X_2^*, \dots, X_n^*$ from F_n . Here, P^* denotes the conditional probability law of $(X_1^*, X_2^*, \dots, X_n^*)$ given $(X_1, X_2, \dots, X_n)$. The rv $a(F_n)$ can be approximated by means of the following Monte Carlo algorithm:

1. Draw a bootstrap sample $X_1^*, X_2^*, \dots, X_n^*$ from F_n and calculate $\hat{\theta}^*, \hat{\sigma}^*$ and

$$T_n^* = |\hat{\theta}^* - \hat{\theta}|/\hat{\sigma}^*$$

2. Independently repeat step 1 some number "B" times, obtaining bootstrap replications $T_{ni}^*, i = 1, 2, \dots, B$. Let $D_{(i)}, i = 1, 2, \dots, B$, denote the order statistics of $T_{ni}^*, i = 1, 2, \dots, B$, then $a(F_n)$ is approximated by $D_{(r)}$, where $r = [B(1 - \alpha)]$, with $[z]$ denoting the largest integer less than or equal to z .

2. Pivotal iteration intervals

A different pivotal construction, which mimics the probability integral approach, was suggested by Beran (1987). Thus if some random variable $Q_0(\hat{\theta}, \theta)$ has c.d.f. $\hat{H}_0$ under sampling from F_n , then

$$Q(\hat{\theta}, \theta) = \hat{H}_0(Q_0(\hat{\theta}, \theta))$$

is very nearly pivotal. If $\hat{H}$ is the c.d.f. of $Q(\hat{\theta}, \theta)$ under sampling from F_n , and if $Q(\hat{\theta}, \theta)$ is monotone in θ , then solutions to

$$\hat{H}(Q(\hat{\theta}, \theta)) = \alpha/2 \text{ and } 1 - \alpha/2$$

define an approximate equal-tailed $(1 - \alpha)$ -confidence interval for θ . The difficulty here is that $\hat{H}$ is based on second-level bootstrapping, i.e. sampling from samples from F_n . The theoretical and numerical properties of this pivotal iteration confidence interval are comparable to those for the bootstrap- t interval discussed above.

3. Percentile intervals

Let $\hat{G}$ denote the c.d.f. of $\hat{\theta}^* = \hat{\theta}(X_1^*, \dots, X_n^*)$, i.e.

$$\hat{G}(t) = P^*(\hat{\theta}^* < t).$$

The *percentile* $(1 - \alpha)$ -confidence intervals are given by

$$(6.1) \quad \theta \in [\hat{G}^{-1}(\alpha/2), \hat{G}^{-1}(1 - \alpha/2)]$$

which can be approximated as follows: Draw B independent bootstrap random samples of size n and calculate $\hat{\theta}_1^*, \hat{\theta}_2^*, \dots, \hat{\theta}_B^*$. If $\hat{\theta}_{(1)}^*, \hat{\theta}_{(2)}^*, \dots, \hat{\theta}_{(B)}^*$ denote the corresponding order statistics, the percentile interval can be approximated by

$$\theta \in [\hat{\theta}_{(r)}^*, \hat{\theta}_{(s)}^*]$$

where the integers r and s are defined by $r = [B\alpha/2]$ and $s = [B(1 - \alpha/2)]$. Johns (1988) shows how the computational demands of this percentile method may be substantially reduced by incorporating importance-sampling techniques.

The *bias-corrected percentile* $(1 - \alpha)$ -confidence intervals are given by

$$(6.2) \quad \theta \in [\hat{G}^{-1}\{\Phi(2z_0 - z(\alpha/2))\}, \hat{G}^{-1}\{\Phi(2z_0 + z(\alpha/2))\}]$$

where Φ is the standard normal c.d.f., $\Phi(z(\alpha/2)) = 1 - \alpha/2$ and

$$z_0 = \Phi^{-1}(\hat{G}(\hat{\theta})).$$

These intervals are an adjustment of the percentile intervals, and they attempt to eliminate the effects of the bias of the bootstrap distribution of $\hat{\theta}^*$. Note that if $\hat{G}(\hat{\theta}) = 0.5$, the median unbiased case, then $z_0 = 0$

and (6.2) reduces to (6.1). Once again, these intervals can be calculated exactly as above, except that we now define

$$r = [B\Phi(2z_0 - z(\alpha/2))],$$

$$s = [B\Phi(2z_0 + z(\alpha/2))], \text{ and}$$

$$\hat{G}(\hat{\theta}) = B^{-1} \sum_{b=1}^B I(\hat{\theta}_b^* < \hat{\theta}).$$

Efron (1982) provides a complete discussion of these two types of confidence intervals. Efron (1987) introduces an improved version of the bias-corrected percentile method, called the *accelerated bias-corrected percentile* method, that incorporates both a bias and skewness correction. These $(1 - \alpha)$ -confidence intervals are defined by

$$(6.3) \quad \theta \in [\hat{G}^{-1}\{\Phi(2z_0 - b(\alpha/2))\}, \hat{G}^{-1}\{\Phi(2z_0 + c(\alpha/2))\}]$$

where

$$b(\alpha/2) = \{z(\alpha/2) - z_0\} / \{1 - a(z_0 - z(\alpha/2))\} + z_0$$

$$c(\alpha/2) = \{z(\alpha/2) + z_0\} / \{1 - a(z_0 + z(\alpha/2))\} - z_0$$

and a is some constant depending on the unknown F . Note that if $a = 0$ then (6.3) reduces to (6.2). Now, suppose $\theta = t(F)$, for some smooth functional t , and $\hat{\theta} = t(F_n)$, then Efron (1987) suggests estimating a by

$$a = \frac{1}{6} \frac{\sum_{i=1}^n U_i^3}{(\sum_{i=1}^n U_i^2)^{3/2}}$$

where U_i is the empirical influence function of $\hat{\theta}$ evaluated at $X_i = x_i$, $i = 1, \dots, n$, i.e.

$$U_i = \lim_{\varepsilon \rightarrow 0} \frac{t((1 - \varepsilon)F_n + \varepsilon\delta_i) - t(F_n)}{\varepsilon}$$

and δ_i is the degenerate distribution function at the point x_i . Instead of letting $\varepsilon \rightarrow 0$ to compute U_i , some small value of ε , for example $\varepsilon = 1/(n + 1)$, is usually selected; see Efron (1982, page 41) for a detailed discussion on this matter.

Remarks

1. The three percentile bootstrap confidence intervals discussed above are in order of increasing generality. All three methods use percentiles of $\hat{G}$ to define the confidence intervals, but differ in which percentiles are used. These procedures arise from transformation theory considerations, and they have the property of invariance under reparameterization. This implies that if the percentile intervals are correct on some transformed scale $\phi = g(\theta)$, then they must also be correct on the original scale θ . The statistician does not need to know the normalizing transformation g , only that it exists. Of these three techniques, the accelerated bias-corrected interval generally performs best.
2. Both the accelerated bias-corrected percentile interval and the bootstrap- t interval are second order accurate; that is, their coverage probabilities differ from the nominal level by only $O(n^{-1})$ instead of $O(n^{-\frac{1}{2}})$ for the standard t , or normal, intervals. This was proved for the mean by Singh (1981) and for more general statistics by Hartigan (1986), Bickel (1987) and Hall (1988a).
3. Hall (1988a) favours the bootstrap- t over the accelerated bias-corrected procedure, provided the standard error estimate $\hat{\sigma}$ is chosen correctly. His decision is based on third-order properties of two-sided confidence intervals.
4. Almost all the published work on bootstrap confidence intervals has focussed on *equal-tailed* two-sided intervals. An equal-tailed $(1 - \alpha)$ -level confidence interval for an unknown parameter θ would be of the form $[\hat{\theta} - \hat{a}, \hat{\theta} + \hat{b}]$, where $\hat{a}$ and $\hat{b}$ are chosen so that

$$P(\theta < \hat{\theta} - \hat{a}) \doteq P(\theta > \hat{\theta} + \hat{b}) \doteq \alpha/2.$$

Hall (1988b) discussed *symmetric* two-sided *bootstrap-t* intervals of the form $[\hat{\theta} - \hat{c}, \hat{\theta} + \hat{c}]$, where $\hat{c}$ is chosen so that

$$P(|\hat{\theta} - \theta| > \hat{c}) \doteq \alpha.$$

Hall (1988b) shows that in quite general circumstances equal-tailed bootstrap- t confidence intervals have coverage error $O(n^{-1})$, that symmetric

bootstrap-t intervals have coverage error $O(n^{-2})$ and that Beran's (1987) pivotal iteration of a symmetric bootstrap-t interval reduces the error to a very low $O(n^{-3})$. It is also proved that symmetric intervals are not necessarily any larger than equal-tailed intervals.

5. The bootstrap-t interval procedure has some drawbacks however. First, it is difficult to compute except for simple linear statistics like the mean. The reason is that a bootstrap computation is needed to calculate $\hat{\sigma}$, the bootstrap estimate of the standard error of $\hat{\theta}$, and thus the entire computation is a double nested bootstrap. A second problem is its lack of invariance. One could apply the bootstrap-t to any transformation of the parameter and hence obtain an interval for the original parameter; the result would depend on the transformation used. Hence the method is not invariant under reparameterization.

6. Bootstrap confidence intervals require more bootstrap replications than do bootstrap standard errors, of the order of $B = 1000$ rather than $B = 50$ or 100.

7. Given the diversity of criteria in choosing a confidence interval procedure, we feel it is unlikely that a single procedure will emerge as a preferred method in all problems.

A modest Monte Carlo study was carried out to illustrate the technique of bootstrap-t for the construction of a 95 % symmetric two-sided confidence interval for the center θ of a symmetric distribution. The β -trimmed mean

$$\hat{\theta} = ([n(1 - 2\beta)] + 1)^{-1} \sum_{i=[n\beta]+1}^{n-[n\beta]} X_{(i)}$$

was considered as an estimator of θ , where $[w]$ is the largest integer less than or equal to w and $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ are the order statistics of $X_1, X_2, \dots, X_n$. Also, we used the natural and well-known estimate of the variance of $\hat{\theta}$, viz.

$$\hat{\sigma}^2 = ([n(1 - 2\beta)] + 1)^{-2} \left\{ \sum_{i=[n\beta]+1}^{n-[n\beta]} (X_{(i)} - \hat{\theta})^2 + [n\beta]((X_{([n\beta]+1)} - \hat{\theta})^2 + (X_{(n-[n\beta])} - \hat{\theta})^2) \right\}$$

The entries in Tables 4 and 5 are estimates based on 2000 independently repeated Monte Carlo trials. For each Monte Carlo trial, the bootstrap estimates were based on $B = 1000$ bootstrap samples of size n . The β -values at the top of each table are the proportions of trimming employed for the trimmed means. The standard normal, $F \sim N(0,1)$, and the standard double exponential, $F \sim DExp(1)$, were considered.

Table 4

Monte Carlo results: $F \sim N(0,1)$

	n=10		n=20	
	$\beta = 0,1$	$\beta = 0,2$	$\beta = 0,1$	$\beta = 0,2$
Coverage	0,963	0,961	0,951	0,941
E(Length)	1,758	2,438	1,002	1,047

Table 5

Monte Carlo results: $F \sim DExp(1)$

	n=10		n=20	
	$\beta = 0,1$	$\beta = 0,2$	$\beta = 0,1$	$\beta = 0,2$
Coverage	0,970	0,961	0,960	0,952
E(Length)	2,182	2,636	1,252	1,114

From the tables we see that the coverage probability of the bootstrap- t behaved satisfactorily. It never differed from its nominal value by more than 2% and was not conservative in only one instance. Note that in the normal case and for each fixed n the expected length of the interval increased as β increased. This is to be expected since in the normal case the sample mean $\bar{X}_n$ is the most precise estimator of θ . As far as the double exponential distribution is concerned, we see that for $n = 10$ the expected length was shorter for $\beta = 0,1$ than $\beta = 0,2$, but when n increased to $n = 20$ the converse was true. These findings are sensible for this heavy tail distribution. Of course, an important question that arises when applying the trimmed mean as an estimator of the point of symmetry θ , is how large β should be chosen. We will show later how the bootstrap can be

used to select the trimming proportion in trimmed means adaptively, i.e. using the data.

6.2 CONFIDENCE BANDS

To end this section we briefly discuss the construction of bootstrap confidence bands for the c.d.f. F for simple random sampling, but do not assume that F is continuous. As before, let F_n be the empirical distribution function of $X_1, X_2, \dots, X_n$ and define the Kolmogorov statistic

$$D_n = \sqrt{n} \sup_x |F_n(x) - F(x)|.$$

Also, for F continuous, let $K_n(\cdot)$ be the c.d.f. of D_n , the (tabled) Kolmogorov distribution. It is well-known that, for all d ,

$$P(D_n \leq d) \geq K_n(d),$$

so we can construct a level $(1 - \alpha)$ -confidence band for F , namely,

$$F_n \pm K_n^{-1}(1 - \alpha)/\sqrt{n}.$$

If F is discrete, this band is conservative. The bootstrap alternative is the following: Let $X_1^*, X_2^*, \dots, X_n^*$ be a bootstrap random sample from F_n and F_n^* the corresponding e.d.f. Set

$$D_n^* = \sqrt{n} \sup_x |F_n^*(x) - F_n(x)|$$

and calculate the bootstrap distribution of D_n^*

$$K_n^*(d) = P^*(D_n^* \leq d).$$

Efron's (1979) Monte Carlo procedure, as described several times above, can be applied to approximate $K_n^*(d)$.

If we let

$$q_n^* = \inf\{d : K_n^*(d) \geq 1 - \alpha\}$$

the bootstrap confidence band is simply

$$F_n \pm q_n^*/\sqrt{n}.$$

Bickel and Freedman (1981) showed that, unless F is a point mass,

$$P(D_n \leq q_n^*) \rightarrow 1 - \alpha$$

so the band is asymptotically correct.

Bickel and Krieger (1989) presented a small Monte Carlo study of the behaviour of bootstrap confidence bands. The distributions F which they considered were (a) uniform on integers 1-10, (b) Poisson(1), (c) binomial(4;0.5), and (d) binomial(9;0.5). These situations were chosen to provide some possibilities of exploration of the effects of (i) *size of carrier*: small finite (c); large finite (a),(d); infinite (b), and (ii) *modality*: multimodal (a); unimodal (b)-(d).

The sample sizes were $n = 20$ and 40 ; the number of Monte Carlo samples was 1000; the number of bootstrap replications per sample was $B = 1000$ and $1 - \alpha = 0.95$. For each F and n the above authors estimated, by averaging across samples, $P(D_n \leq q_n^*)$, the probability of coverage of the bootstrap band, and $P(D_n \leq K_n^{-1}(1 - \alpha))$, the probability of coverage of the conservative Kolmogorov-Smirnov band. They also estimated q_n , the actual $(1 - \alpha)$ quantile of the cumulative distribution function of D_n , and $E(q_n^*)$ and the percentage differences

$$(E(q_n^*) - q_n)/q_n$$

and

$$(K_n^{-1}(1 - \alpha) - q_n)/q_n.$$

Bickel and Krieger (1989) summarized their conclusions (see Tables 6 and 7) as follows:

1. The coverage probability of the bootstrap behaved satisfactorily. It never differed from its nominal value by more than 1% and was not conservative in only three instances, binomial(4;0.5) for $n = 20$ and $n = 40$ and Poisson for $n = 20$.

2. The percentage differences between the expected bootstrap quantile and the actual was negligible. The conservative band was consistently between 20% (for $n = 20$) and 30% (for $n = 40$) too long.

3. The risk of the bootstrap in terms of coverage probability decreased with n and its advantage over the conservative band increased with n .

Table 6
Coverage probabilities for the 95 % bands

Distribution	B		C	
	n=20	n=40	n=20	n=40
a	0,96	0,96	0,99	0,98
b	0,94	0,95	0,99	0,99
c	0,94	0,94	0,98	0,99
d	0,95	0,96	0,99	0,99

Note: B, bootstrap; C, conservative.

Table 7
Quantiles for the 95 % bands

		Distribution			
		a	b	c	d
A	n=20	0,25	0,23	0,24	0,25
	n=40	0,18	0,16	0,16	0,18
B	n=20	0,25	0,22	0,22	0,24
	n=40	0,18	0,16	0,16	0,17
C	n=20	0,29	0,29	0,29	0,29
	n=40	0,21	0,21	0,21	0,21
(B-A)/A	n=20	0,00	0,00	-0,06	-0,03
	n=40	0,00	0,00	0,00	-0,02
(C-A)/A	n=20	0,18	0,27	0,24	0,18
	n=40	0,20	0,33	0,29	0,20

Note: A, actual; B, bootstrap; C, conservative.

Remarks

I. Bickel and Krieger (1989) also investigated the analogous problem of finding bootstrap confidence bands for F for a more complicated situation, namely stratified random sampling as discussed in Bickel and Freedman (1984).

2. Jhun (1988) discusses the problem of constructing confidence bands for *probability density functions* based on bootstrap methods. He shows that these bands have the correct asymptotic level and their small-sample behaviour is very satisfactory.

7. BOOTSTRAP ADAPTIVE STATISTICAL PROCEDURES

Numerous estimators in different problems, ranging from the estimation of a finite dimensional location parameter to the infinite dimensional estimation of a density, are only specified up to a constant. In most cases, the optimal choice of this constant is a function of the unknown F and the sample size. Using a fixed value will lead to good or bad estimation depending on the particulars of the sample at hand. Consequently, the unknown constant is often chosen *adaptively*, that is, using the data (see, e.g. Jaeckel, 1971).

The bootstrap can successfully be used to construct adaptive statistical procedures with good finite-sample results and good asymptotic properties. In his dissertation, Léger (1988) shows, among others, how the bootstrap can be used to adaptively select the trimming proportion in trimmed means. His idea is essentially the following:

Let $\hat{\theta} = \hat{\theta}(\beta)$ be the β -trimmed mean as defined above and denote the bootstrap estimate of the standard error of $\hat{\theta}$ by $\hat{\sigma}(\beta)$, where

$$\hat{\sigma}(\beta) = [\text{Var}_* \hat{\theta}^*]^{\frac{1}{2}}$$

The calculation of $\hat{\sigma}(\beta)$ was discussed in Section 3. Now choose $\hat{\beta}_n$ which minimizes $\hat{\sigma}(\beta)$. The bootstrap adaptive trimmed mean is then $\hat{\theta}(\hat{\beta}_n)$.

Léger (1988) proves that $\hat{\theta}(\hat{\beta}_n)$ is asymptotically normally distributed with an asymptotic variance equal to the smallest asymptotic variance among the class of trimmed means. He also shows that the coverage probability of bootstrap confidence intervals based on $\hat{\theta}(\hat{\beta}_n)$ converges, as $n \rightarrow \infty$, to the claimed level of confidence. Note that such a confidence interval involves a *double* bootstrap, and hence, is computationally very intensive. Simulation results demonstrate that $\hat{\theta}(\hat{\beta}_n)$ has very good *finite* sample properties, even for samples of size as small as 10! Finally, Léger presents a short discussion about the possible extensions to more interesting and

important problems, such as regression and density estimation.

8. THE MODIFIED BOOTSTRAP

Suppose $T_n(\mathbf{X}_n; F)$ is some specified random variable of interest, depending upon the underlying c.d.f. F . As stated in Section 2, the bootstrap method consists of approximating the sampling distribution of $T_n(\mathbf{X}_n; F)$ under F by the bootstrap distribution of $T_n(\mathbf{X}_n^*; F_n)$ under F_n , where $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$ denotes a random sample of size n from F_n , i.e.

$$(8.1) \quad P(T_n(\mathbf{X}_n; F) \in B) \approx P^*(T_n(\mathbf{X}_n^*; F_n) \in B)$$

for any Borel set B , and P^* denotes the conditional probability law of $T_n(\mathbf{X}_n^*; F_n)$ given $\mathbf{X}_n$.

Singh (1981), and Bickel and Freedman (1981) show that approximation (8.1) is asymptotically (as $n \rightarrow \infty$) valid in a large number of situations, including t-statistics, the empirical and quantile processes, and von Mises functionals. However, they also provide some counter-examples to show where the bootstrap apparently fails. These examples are for pivotal quantities based on (1) a U-statistic, (2) extreme order statistics and (3) spacings of the observations $\mathbf{X}_n$.

These counter-examples can be mended by introducing the "modified bootstrap". This boils down to replacing $T_n(\mathbf{X}_n^*; F_n)$ in (8.1) by $T_m(\mathbf{X}_m^*; F)$, for some suitable choice of the bootstrap sample size m in terms of n .

Now, consider the following "counter-example" to Efron's (1979) bootstrap method given by Bickel and Freedman (1981): Let $X_1, X_2, \dots, X_n$ be independent, identically distributed random variables from the uniform distribution F on the interval $(0, \theta)$. Using the natural pivot $n(\theta - X_{(n)})/\theta$, where $X_{(i)}$ denotes the i -th order statistic, they observe that

- (i) $n(\theta - X_{(n)})/\theta$ tends to a limiting exponential distribution; and
- (ii) with probability one, the conditional distribution of $n(X_{(n)} - X_{(n)}^*)/X_{(n)}$ does not have a weak limit!

Here $X_{(i)}^*$ denotes the i -th order statistic of the bootstrap sample $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$. Contrary to this finding, Swanepoel (1986b) provides the following result for uniformly $(0, \theta)$ distributed random variables:

Let $m = m(n)$ be an integer such that

$$An^\epsilon \leq m(n) = o(n^{(\epsilon+1)/2}/(\log n)^{\frac{1}{2}})$$

as $n \rightarrow \infty$, for some constants $A > 0$ and $0 < \epsilon < 1$. Then along almost all sample sequences $X_1, X_2, \dots$, the conditional law of

$$m(X_{(n)} - Y_m)/X_{(n)}$$

given X_n converges weakly to Z , where Z is a random variable with the exponential density $\exp(-z)$, for $z \geq 0$, and $Y_m = \max(X_1^*, X_2^*, \dots, X_m^*)$. The other counter-examples of Bickel and Freedman (1981) can be mended similarly.

The following example of bootstrap point estimation provides some additional evidence that there is frequently advantage to be gained from choosing the bootstrap sample size m different from n :

Let $X_1, X_2, \dots, X_n$ be independent, identically distributed *positive* random variables with expectation $\mu > 0$ and finite variance $\sigma^2 > 0$. Suppose

$$T_n(\mathbf{X}_n; F) = \sum_{i=1}^n X_i$$

and we wish to estimate

$$ET_n(\mathbf{X}_n; F) = n\mu.$$

The modified bootstrap estimator of $ET_n(\mathbf{X}_n; F)$ is

$$E_* T_m(\mathbf{X}_m^*; F_n) = m\bar{X}_n$$

where E_* denotes expectation over the conditional law of $\mathbf{X}_m^*$ given $\mathbf{X}_n$. It is easy to check that the mean squared error of $m\bar{X}_n$, i.e. $E(m\bar{X}_n - n\mu)^2$, is minimized by

$$m = n^2 \mu^2 / (n\mu^2 + \sigma^2) < n.$$

It is interesting to note that if in the definition of m, μ and σ^2 are replaced by the sample mean $\bar{X}_n$ and the sample variance s_n^2 respectively, the modified bootstrap estimator of μ reduces to

$$(\hat{m}/n)\bar{X}_n = \bar{X}_n^3 / (\bar{X}_n^2 + s_n^2/n)$$

which is the well-known adaptive shrunken mean estimator of μ , discussed by many authors, see for example page 1430 of Oehlert (1985).

Bretagnolle (1983) and Athreya (1985, 1987) proved that when F is in the domain of attraction of a nonnormal stable law and hence has infinite variance, the naive bootstrap ($m \equiv n$) of the sample mean $\bar{X}_n$ fails, in the sense that the limiting distributions of $\bar{X}_n$ and its bootstrap version are quite different, the latter one being a random probability distribution. These authors concluded that, in this case, the bootstrap of the mean can work only if $m/n \rightarrow 0$.

9. POINT ESTIMATION

Motivated by the results in Section 8, Swanepoel (1988) discusses the problem of constructing estimators for a parameter θ which can be written in the form

$$\theta = \psi(F)$$

for some suitable functional ψ . The nonparametric maximum likelihood estimator of θ is $\psi(F_n)$. However, sometimes this representation of θ does not lead to a sensible estimator $\psi(F_n)$. For example, if $\theta = f(x)$, the density of F at some point x , then

$$\theta = \psi(F) = F'(x)$$

and the estimate $\psi(F_n)$ is useless in that $F'_n(x)$ is either 0 or does not exist. How can this situation be remedied?

Suppose $\psi(F)$ can be approximated by a sequence of functionals, namely

$$\psi_m(F) \approx \psi(F)$$

with the approximation becoming increasingly accurate as $m \rightarrow \infty$. It then seems reasonable to take $\psi_m(F_n)$ as an estimator of θ , with $m = m(n)$ suitably chosen.

Suppose $T_m(X_1, \dots, X_m)$ is some (known) estimator of θ . Two possible choices of $\psi_m(F)$ are

1. $\psi_m(F) = E(T_m(X_1, \dots, X_m))$
2. $\psi_m(F) = \text{med}(T_m(X_1, \dots, X_m))$

where "med" stands for the median of a random variable.

To illustrate the above bootstrap estimation technique, consider the following two examples.

Example 1. Estimation of the p-quantile

Suppose we wish to estimate $F^{-1}(p)$. Put $r = [(m+1)p]$ and let $T_m(X_1, \dots, X_m) = X_{m,r}$, where $X_{m,1} \leq X_{m,2} \leq \dots \leq X_{m,m}$ are the order statistics of $X_1, \dots, X_m$. Define $\psi_m(F) = E(X_{m,r})$, then it is easy to prove that $\psi_m(F_n)$ reduces to

$$\psi_m(F_n) = \sum_{j=1}^n \left\{ \int_{(j-1)/n}^{j/n} b_m(x) dx \right\} X_{(j)}$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ are, as before, the order statistics of $X_1, \dots, X_n$ and $b_m(x)$ denotes the beta $(r, m-r+1)$ density function. This is the quantile estimator studied by Kaigh (1983).

Example 2. Estimation of the mean

We now briefly discuss the derivation of a new estimator for the population mean $\theta = E(X_i)$, which according to Huber (1981), for example, is a difficult parameter to estimate well, especially when F is asymmetric. Now, let

$$\psi_m(F) = \text{med}(\bar{X}_m), \quad \bar{X}_m = m^{-1} \sum_{i=1}^m X_i$$

and hence the new estimator becomes

$$\psi_m(F_n) = \text{med}_*(\bar{X}_m^*), \quad \bar{X}_m^* = m^{-1} \sum_{i=1}^m X_i^*$$

where $X_1^*, X_2^*, \dots, X_m^*$ is a random bootstrap sample of size m from F_n .

For fixed n , it follows that

$$\psi_1(F_n) = \text{med}(X_1, \dots, X_n)$$

$$\psi_2(F_n) = \text{the Hodges-Lehmann estimator}$$

$$\lim_{m \rightarrow \infty} \psi_m(F_n) = \bar{X}_n.$$

Swanepoel (1988) suggests using $m = n/3$, which is a good choice from a mean squared error point of view, as is also confirmed by the Monte

Carlo studies below. He also proves that $\psi_m(F_n)$ is strongly consistent and asymptotically normally distributed as $n \rightarrow \infty$. Furthermore, in the robust terminology, the estimator is "B-robust at F " for each fixed n .

Monte Carlo experiments based on $B=1000$ bootstrap replications in each of 2000 trials were performed. Tables 8 and 9 display the ratios

$$MSE(\bar{X}_n)/MSE(\psi_m(F_n))$$

for $n = 10, 30, 50$ and the following distributions

1. contaminated normal, $F \sim (1-p)N(0,1) + pN(0,3)$, for $p = 0\%, 5\%, 10\%$ and 20% .

2. F -distribution with 8 and 5 degrees of freedom, $F \sim F_{8,5}$.

The results in Tables 8 and 9 show that $\psi_m(F_n)$ performed better than $\bar{X}_n$ in that its mean squared error was always lower than that of $\bar{X}_n$, except in the normal case ($p = 0\%$) where it was at least 97% as efficient as $\bar{X}_n$. This is to be expected, since $\bar{X}_n$ is admissible in the normal case. In this case the choice amongst the three alternatives of m considered did not seem to be critical. However, in all other cases $m = n/3$ was the best and this choice yielded an estimator which outperformed $\bar{X}_n$ substantially. The improvement in mean squared error ranged from 5% to 177%!

Remarks

The above suggested method of constructing estimators for $\theta = \psi(F)$ by means of the bootstrap gives rise to a multitude of possible estimators, each corresponding to a different choice of functional sequence $\{\psi_m(F)\}$. There is still a lot of work to be done in this area, especially for small samples. For example, the issue of best functional sequence to be used in each case needs to be addressed. Swanepoel (1986a) discussed the proposed bootstrap estimation technique in the context of density estimation.

Table 8

M.S.E. ratios: $F \sim (1-p)N(0,1) + pN(0,3)$

		n=10	n=30	n=50
p=0,00	m=n/3	0,974	0,994	0,997
	m=2n/3	0,996	0,995	1,000
	m=n	0,999	0,996	1,000
p=0,05	m=n/3	1,147	1,097	1,050
	m=2n/3	1,071	1,046	1,024
	m=n	1,034	1,029	1,015
p=0,10	m=n/3	1,254	1,125	1,086
	m=2n/3	1,116	1,059	1,040
	m=n	1,049	1,034	1,024
p=0,20	m=n/3	1,296	1,121	1,076
	m=2n/3	1,120	1,055	1,032
	m=n	1,059	1,033	1,022

Table 9

M.S.E. ratios: $F \sim F_{8,5}$

	n=10	n=30	n=50
m=n/3	2,775	1,976	2,416
m=2n/3	2,365	1,358	1,069
m=n	1,039	1,084	1,079

10. GENERAL REMARKS

Since the introduction of the bootstrap procedure by Efron (1979; 1982), there has been a fast-growing literature on the topic. Empirical evidence has suggested that the bootstrap usually performs very well. Simultaneously, there have been attempts to provide theoretical justification as to why this method performs well. These results provide an insight into the working of the bootstrap procedure. In addition to the papers discussed

under the above headings, the following papers are important in the theory and application of the bootstrap.

Romano (1988b) discusses the application of the bootstrap to obtain critical values in a fairly broad class of hypothesis-testing problems. The resulting tests are shown to have the correct asymptotic level under minimal assumptions. The power of the bootstrap method is apparent in these problems because, unlike many statistical problems where the proposed statistics have Gaussian or chi-squared limiting distributions, the test statistics considered by Romano (1988b) do not possess a tractable asymptotic distribution from which critical values could be approximated.

Boos and Brownie (1989) describe how bootstrap methods can be used to get valid critical values for tests of homogeneity of scale but with special emphasis on tests on variances. In fact, an important feature of the proposed resampling methods is their flexibility with respect to the scale parameter of interest. When variances are of interest, a statistic based on sample variances is bootstrapped. If a different scale parameter is preferred, then a statistic based on estimates of this parameter may be used. This is in contrast to methods based on analysis of variance (ANOVA), for which valid tests are obtained across distributions when absolute deviations from the median are used but not when squared deviations from the mean are used [compare Lev 1: med and Lev 4 in Tables 5 and 6 of Conover et al. (1981)]. Boos and Brownie (1989) also conclude that bootstrapping studentized test statistics results in better α levels than bootstrapping unstudentized versions. This agrees with second-order asymptotic theory for other situations (e.g., Babu and Singh 1983; Helmers 1987).

The bootstrap often provides "better than normal" estimates of distribution functions of studentized statistics. This is proved, for example, by Singh (1981) for Student's statistic. The bootstrap is also shown by Bhattacharya and Qumsiyeh (1989) to be more accurate than even the two-term *empirical* Edgeworth expansion, thus strengthening the claim of superiority of the bootstrap over the normal approximation method. The two methods were compared not only with respect to bounded bowl-

shaped loss functions but also with respect to squared error loss and, more generally, in L^p -norms.

Another significant paper which deserves to be mentioned is by Bose (1988) who proves that the distribution of least-squares estimates in stationary autoregressive processes can be bootstrapped with accuracy $o(n^{-\frac{1}{2}})$ almost surely, thereby improving the normal approximation error of $O(n^{-\frac{1}{2}})$. This result is similar to that obtained by Singh (1981) for the sample mean in the i.i.d. situation. Basawa, et al. (1989) establish the asymptotic validity of the bootstrap for the least squares estimate of the parameter of a *nonstationary* (explosive) first-order autoregressive process.

Gill (1989) analyzes nonlinear functions of the empirical distribution function, such as quantiles and the Mann-Whitney statistic by way of the delta method, and shows that the bootstrap "works" for Hadamard differentiable functions. Wellner's (1989) discussion on this paper indicates how Gill's methods extend to functionals of empirical measures more generally. Csörgö and Mason (1989) develop the complete bootstrapped parallel to the asymptotic theory of weighted empirical and quantile processes. Utilizing this parallel theory, they present a general body of techniques to establish the asymptotic validity of the bootstrap method of constructing confidence bands for statistical functions. These techniques are demonstrated to be applicable to the construction of asymptotic bootstrap confidence bands for a variety of concrete functions. In their paper, Giné and Zinn (1990) prove that the bootstrap works for general empirical measures, giving very precise "if and only if" theorems.

Efron's (1979) bootstrap has a Bayesian counterpart called the Bayesian bootstrap introduced by Rubin (1981). In a recent article, Lo (1987) shows that, for a variety of popular functionals, the Bayesian bootstrap and the bootstrap are first-order asymptotically equivalent in the sense that, for almost all sample sequences and subject to proper centerings and the $n^{\frac{1}{2}}$ -scaling, they achieve the same limiting conditional distribution. This result indicates that the uses of the Bayesian bootstrap and the bootstrap can be interchanged, at least in the first-order asymptotic sense. An implication is that a frequentist can use the Bayesian bootstrap

to approximate the sampling distribution of a statistic. In addition, Weng (1989) shows that the Bayesian bootstrap approximation to the posterior distribution of the unknown mean based on a Dirichlet process prior is more accurate than both the standard normal approximation discussed in Lo (1987) and the bootstrap approximation suggested by Efron (1982, page 82). It is also shown by Weng (1989) that the Bayesian bootstrap approximation to the sampling distribution of the sample average is only as good as the standard normal approximation, yet not as accurate as the one based on the bootstrap method (Singh, 1981).

The bootstrap method for a *finite* population was introduced and discussed by Gross (1980), Bickel and Freedman (1984) and Chao and Lo (1985). Recently, Lo (1988) also introduced a Bayesian bootstrap for a finite population and discussed its small-sample distributional properties. He shows that the Bayesian bootstrap and the frequentist bootstrap for a finite population are first-order asymptotically equivalent.

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